

ECE 647: Lecture 20

Cross-Layer formulation:

In an optimization approach, it is not difficult to incorporate controls at multiple layers into a unified optimization problem.

- Physical layer:
 - Power Control, Water filling
 - Uses rate-power function
- MAC:
 - Scheduling
- Network Layer:
 - Multi-path routing
 - Node-balance equation
- Transport Layer
 - Utility maximization
 - Revenue maximization

So we have various combinations. Key consideration is

- Convexity
- Distributed / decomposed solution

One way of putting all together:

$$\begin{aligned} & \max \sum_s U_s(X_s) - \text{utility/congestion control} \\ & \text{subject to } X_s = \sum_j X_{sj} - \text{routing/load-balancing,} \end{aligned} \quad (1)$$

where X_{sj} is the rate of user s on path j .

$$\sum_{s_j} H'_{sj} X_{sj} \leq r_l \quad (2)$$

where $H'_{sj} = 1$ if path ' j ' of User ' s ' uses path ' l '

$$r_l \leq \sum_{k=1}^K g_k r_l^{(k)}, \forall l \quad (3)$$

where, g_k is given probability of channel state ' k ', with $\sum_{k=1}^K g_k = 1$.
 K channel states and each may use a different decision (schedule k).

$$r_l^{(k)} \leq g_l^{(k)} \left(\vec{p}^k \right) \quad (4)$$

- power control adaptive coding/ modulation,

where $g_l^{(k)}$ is rate power function and $\vec{p}^k$ is power assignment for schedule k .

Other formulations

- replace H'_{sj} by node-balance equation
- Use more than one schedule per channel state
- The set of schedules can be very large.

In general, this may not be a convex problem.

- The function $g_l^{(k)}(\cdot)$ may not be concave.
- For example, we may use Shannon's capacity

$$g_l^k \left(\vec{p}^k \right) = W \cdot \log \left(1 + \frac{p_l^k \cdot g_{ll}^k}{\sum_{h \neq l} p_l^k \cdot g_{hl}^k + N_l} \right)$$

- However, the constraint $r_l^{(k)} \leq g_l^{(k)} \left(\vec{p}^k \right)$ is not convex in general!

Special cases that are Convex

1 Fixed Link Rate $\rightarrow$ Pure Power Control

- We want a certain target rate to be reached for $r_l^{(k)}$, which is given before hand
 - This target rate may be independent of the channel k
 - Good channel $\rightarrow$ Smaller power
 - Bad channel $\rightarrow$ Higher power

- In order for

$$W \cdot \log \left(1 + \frac{p_l^k \cdot g_{ll}^k}{\sum_{h \neq l} p_l^k \cdot g_{hl}^k + N_l} \right) \geq r_l^k \doteq W \cdot \log(\gamma_l),$$

we just need,

$$1 + \frac{p_l^k \cdot g_{ll}^k}{\sum_{h \neq l} p_l^k \cdot g_{hl}^k + N_l} \geq \gamma_l \quad (\gamma_l > 1),$$

$$p_l^k \cdot g_{ll}^k \geq (\gamma_l - 1) \left(\sum_{h \neq l} p_l^k \cdot g_{hl}^k + N_l \right) \quad (5)$$

This is the formulation for power control problems

$\min \sum_l p_l^k$ subject to Equation (5), which is a linear equation.

2 No Interference

- Link rate a function of its own power

$$g_k(p_l) = W \cdot \log \left(\frac{p_l \cdot g_{ll}}{N_l} \right)$$

which is a concave function.

eg. Water filling for fading channels

3 No power control

- Link rate is fixed
- Can still add the possibility of scheduling power

Example: Opportunistic scheduling for multi-user systems with fading

4 Approximation (e.g High SINR)

Water filling for fading channels

- Single channel per link per user

$$\sigma = \log \left(1 + \frac{gP}{N} \right) \quad (6)$$

where g is the Channel gain, P is the transmission power and N is the background noise with no interference.

- But the channel gain changes in time. Assume that the channel gain can be one of the K values, $g_1, g_2, \dots, g_k$

$$q_k = Pr(g = g_k)$$

- How do we control the transmission strategy

- Specifically, suppose our assigned power must be no greater than P_0 on average.
- Use same power for all channel gains?

$$\sum_k g_k \log \left(1 + \frac{g_k P_0}{N} \right) \quad (7)$$

- Or, we can vary the transmission power at each channel gain.

- Intuition: If the average transmission power is limited, then we may want to allocate more power for bad channels, so that the total transmission rate can be bigger.
- Note that this is opposite to the intuition for power control!
 - May produce larger delay
- Let P_k be the transmission power when the channel gain is g_k
 - Average power is $\sum_{k=1}^M g_k p_k$
 - Total Transmission rate is

$$\sum_{k=1}^M g_k \log \left(1 + \frac{g_k p_k}{N} \right) \quad (8)$$

$$\max_{p_1, p_2, \dots, p_m} \sum_{k=1}^M g_k \log \left(1 + \frac{g_k p_k}{N} \right) \quad (9)$$

$$\text{sub to} \quad \sum_{k=1}^M g_k p_k \leq P_0, p_k \geq 0$$

- which is a Convex problem!

Solution:

- Since the objective is increasing in p_k , the solution must be at

$$\sum_{k=1}^M g_k p_k = P_0$$

- Assume in addition that $P_k > 0$
- The normal vector is $[g_1, g_2, \dots, g_m]$
- The gradient of

$$-\sum_k g_k \log \left(1 + \frac{g_k p_k}{N} \right) \text{ is}$$

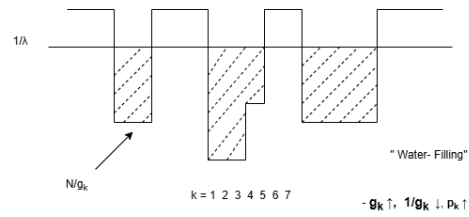
$$\left[-\frac{g_k \cdot \frac{g_k}{N}}{1 + \frac{g_k p_k}{N}} \right]_{k=1, \dots, M}$$

$$\Rightarrow \frac{g_k \cdot \frac{g_k}{N}}{1 + \frac{g_k p_k}{N}} = \lambda g_k$$

$$\Rightarrow 1 + \frac{g_k}{N} \cdot p_k = \frac{g_k}{N} \cdot \frac{1}{\lambda}$$

$$\Rightarrow p_k = \left(\frac{1}{\lambda} - \frac{N}{g_k} \right)$$

- λ is chosen such that $\sum p_k = P_0$



- What if $\frac{1}{\lambda} - \frac{N}{8k} < 0$? (Wait when we discuss duality)